

9 Large-Scale Agentic Web Stress Test

Design. We test whether lineage-first distribution keeps knowledge useful at population scale, on a simulated agentic web of $N \in \{10^2, 10^3, 10^4, 10^5\}$ agents in teams, failure domains, organisations and regions. Every evidence item has a home agent and descends from exactly one origin; family sizes are heavy-tailed and $\sim 35\%$ of claims have a single origin, so replica count and independent support come apart by construction. We score 127,374 labelled questions per seed at $N = 10,000$ (single-hop, multi-hop, temporal, revision-sensitive, reconstruction, contradiction, evidence-verification, strategic ranking). Equal-budget systems publish an identical 4 replicas per claim and every probe attempt is charged as a message, dead hosts included. Failures escalate from 10–90% churn to correlated loss of whole organisations, to a **white-box adversary** that knows each system's own placement and removes the hosts of its scarcest surviving independent evidence, to partitions with independent updates, to corruption of nodes or of origin sources. 30 seeds, paired by seed, Holm-corrected; 50,700 recorded runs in total.

Resilience. Under the white-box attack at $N = 10,000$, knowledge survival is *statistically indistinguishable* across the equal-budget policies (B6–B3 +0.000, $p = 0.59$; Fig. 1c). What separates them is the quality of what survives: independent-support survival +0.046 ($d_z = 21.3$), contradiction F1 +0.155, task accuracy +0.010, all Holm $p < 1e-6$. The gap widens under origin-source corruption, where a corrupted origin's false value is inherited by all of its copies (+0.019 accuracy). Broad replication survives better (0.870) but at 194 replicas per claim, 48 $\times$ the storage: on the cost frontier the margin is bought with capacity, not with a better use of it. The effect is flat in N from 10^2 to 10^5 .

Continual questioning, and what does not hold. At identical storage — the base budget is reduced by exactly the replicas questioning adds — questioning raises accuracy by +0.189 ($d_z = 37.2$) for a 1.8 $\times$ increase in messages, and drives post-partition reconciliation. But the identical machinery on a count-based system gains *more* (+0.245): continual discovery is largely orthogonal to lineage, not a consequence of it. Three further negatives: in a world where every claim has one origin the lineage fabric and random replication are identical; almost all of the questioning gain is re-verification rather than acquiring new independent support; and targeting questions by lineage fragility is no better than targeting them at random. Lineage awareness improves the *evidential* quality of what survives at matched cost; it does not, on its own, make more of it survive.

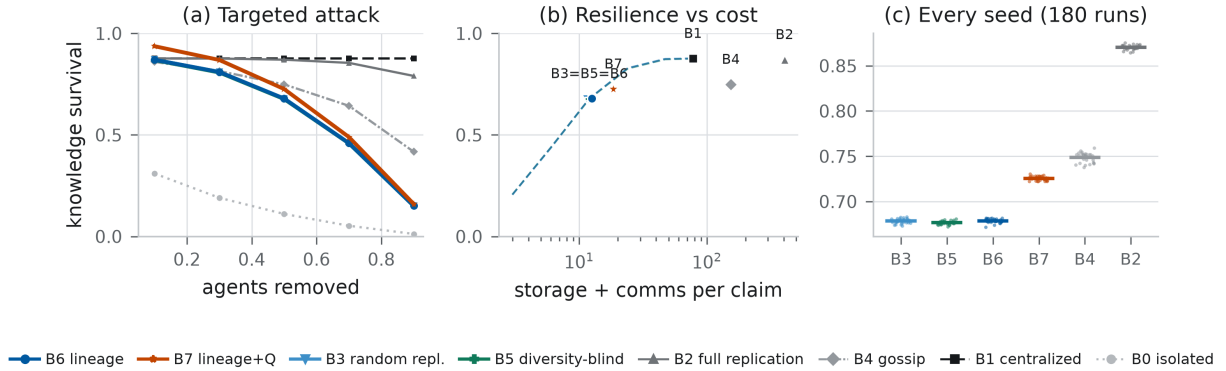


Figure 1: (a) Knowledge survival as a white-box adversary removes agents ($N = 10,000$, 30 seeds, bands 95% CI); axes span the full $[0, 1]$ range. (b) Survival against storage + communication cost per claim at 50% removal; dashed line sweeps the storage budget $k \in \{1...32\}$, markers dodged where systems coincide. (c) The same 50% condition with every seed drawn as one point on a zoomed axis: the three equal-budget policies overlap, while questioning and the high-cost systems separate cleanly.

	KS	acc.	ISS	contr. F1	ECE*	storage	msgs
B2 full replication	0.870	0.597	0.993	0.895	0.006	194	210
B4 gossip	0.749	0.472	0.691	0.747	0.007	48	105
B3 random replication	0.678	0.441	0.504	0.351	0.006	4	8
B5 diversity-blind	0.677	0.429	0.437	0.000	0.005	4	8
B6 lineage-aware fabric	0.679	0.451	0.550	0.506	0.008	4	8
B7 lineage + questioning	0.726	0.594	0.550	0.398	0.004	4	15
B3Q questioning, no lineage	0.740	0.628	0.507	0.243	0.004	4	16

Table 1: The 50% white-box attack at $N = 10,000$ (30 seeds, paired). KS counts knowledge that is both recoverable and answered correctly; ISS is the fraction of original independent evidence paths surviving in the system's own stored set; ECE* is calibration error after held-out recalibration; storage counts published replicas per claim and messages counts publication, questioning and every probe attempt. B3, B5, B6 and B7 share one storage budget; B2 and B4 do not, and their cost columns show what their margin is bought with. Full grid, all conditions and the negative-result table are in the supplement.